

Quantum Information Theory

Lesson 6: Quantum Data Processing

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Contents

- ▶ Umegaki Divergence
- ▶ Bistochastic Channels
- ▶ Reduced States
- ▶ No-Communication Theorem
- ▶ Quantum Mutual Information
- ▶ Holevo Theorem
- ▶ Conditional von Neumann Entropy
- ▶ Schmidt Decompositions
- ▶ Entropy of Entanglement
- ▶ Uncertainty Relations with Quantum Memory
- ▶ Tripartite Uncertainty Relations
- ▶ Quantum Key Distribution: BB84 Security Proof

Umegaki Divergence

While the classical relative information is called the KL divergence, the quantum relative information is called the Umegaki divergence,

$$D(\rho\|\sigma) = -\text{Tr}(\rho(\log \sigma - \log \rho)).$$

Note that if there is $|\psi_0\rangle \in \ker \sigma$ yet $|\psi_0\rangle \perp \ker \rho$ then extending it up to an eigenbasis of $\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|$ we find

$$-\langle \psi_0 | \rho(\log \sigma - \log \rho) | \psi_0 \rangle = -p_0 \langle \psi_0 | \log \sigma | \psi_0 \rangle = \infty,$$

so the right side is not well defined, thus we then set $D(\rho\|\sigma) = \infty$. Otherwise, the eigendecomposition $\sigma = \sum_j q_j |\phi_j\rangle \langle \phi_j|$ produces

$$\begin{aligned} D(\rho\|\sigma) &= \sum_i p_i (\log p_i - \sum_j |\langle \psi_i | \phi_j \rangle|^2 \log q_j) \\ &\geq \sum_i p_i (\log p_i - \log \sum_j q_j |\langle \psi_i | \phi_j \rangle|^2) = D_{KL}(\mathbf{p}\|\mathbf{r}) \geq 0, \end{aligned}$$

and for equality we need all $|\langle \psi_i | \phi_j \rangle|^2 \in \{0, 1\}$ i.e. $|\psi_i\rangle = |\phi_{\tau(i)}\rangle$ for a permutation τ , thus $r_i = q_{\tau(i)}$, and we need $\mathbf{D}_{KL}(\mathbf{p}\|\mathbf{r}) = 0$ i.e. $\mathbf{p} = \mathbf{r}$, thus actually equality is achieved iff $\rho = \sigma$.

The Quantum Data Processing Inequality

It is possible (but very technical) to prove joint convexity i.e.

$$D\left(\sum_i p_i \rho_i \parallel \sum_i p_i \sigma_i\right) \leq \sum_i p_i D(\rho_i \parallel \sigma_i).$$

Thus, for a mixed unitary channel $\mathcal{E}(\rho) = \sum_i p_i \mathbf{U}_i \rho \mathbf{U}_i^\dagger$, we have

$$D(\mathcal{E}(\rho) \parallel \mathcal{E}(\sigma)) \leq \sum_i p_i D(\mathbf{U}_i \rho \mathbf{U}_i^\dagger \parallel \mathbf{U}_i \sigma \mathbf{U}_i^\dagger) = \sum_i p_i D(\rho \parallel \sigma) = D(\rho \parallel \sigma),$$

so for any quantum channel $\mathcal{E}(\rho_A) = \text{Tr}_E(\mathbf{U}_{AE}(\rho_A \otimes |0\rangle \langle 0|_E) \mathbf{U}_{AE}^\dagger)$,

$$\begin{aligned} D(\rho_A \parallel \sigma_A) &= D(\rho_A \otimes |0\rangle \langle 0|_E \parallel \sigma_A \otimes |0\rangle \langle 0|_E) \\ &= D(\mathbf{U}_{AE}(\rho_A \otimes |0\rangle \langle 0|_E) \mathbf{U}_{AE}^\dagger \parallel \mathbf{U}_{AE}(\sigma_A \otimes |0\rangle \langle 0|_E) \mathbf{U}_{AE}^\dagger) \\ &\geq D(\mathbf{I}_A \otimes \Delta_{1E}(\mathbf{U}_{AE}(\rho_A \otimes |0\rangle \langle 0|_E) \mathbf{U}_{AE}^\dagger) \parallel \mathbf{I}_A \otimes \Delta_{1E}(\mathbf{U}_{AE}(\sigma_A \otimes |0\rangle \langle 0|_E) \mathbf{U}_{AE}^\dagger)) \\ &= D(\text{Tr}_E(\mathbf{U}_{AE}(\rho_A \otimes |0\rangle \langle 0|_E) \mathbf{U}_{AE}^\dagger) \otimes \frac{\mathbf{I}_d}{d} \parallel \text{Tr}_E(\mathbf{U}_{AE}(\sigma_A \otimes |0\rangle \langle 0|_E) \mathbf{U}_{AE}^\dagger) \otimes \frac{\mathbf{I}_d}{d}) \\ &= D(\mathcal{E}(\rho_A) \parallel \mathcal{E}(\sigma_A)), \end{aligned}$$

known as the quantum data processing inequality.

Bistochastic Channels Revisited

In particular, notice that

$$\begin{aligned} 0 &\leq D(\rho \| \mathbf{I}_d/d) - D(\mathcal{E}(\rho) \| \mathcal{E}(\mathbf{I}_d/d)) \\ &= -\text{Tr}(\rho(\log \mathbf{I}_d/d - \log \rho)) + \text{Tr}(\mathcal{E}(\rho)(\log \mathcal{E}(\mathbf{I}_d/d) - \log \mathcal{E}(\rho))) \\ &= H(\mathcal{E}(\rho)) - H(\rho) + (1 - \text{Tr} \mathcal{E}(\rho) \log \mathcal{E}(\mathbf{I}_d)) \log d, \end{aligned}$$

and since bistochastic channels $\mathcal{E}$ have $\mathcal{E}(\mathbf{I}_d) = \mathbf{I}_d$ we find that for such channels $H(\mathcal{E}(\rho)) \geq H(\rho)$ i.e. they cannot decrease entropy. Conversely, such channels must have $\mathcal{E}(\mathbf{I}_d/d) = \mathbf{I}_d/d$, so **a channel cannot decrease entropy iff it is bistochastic.**

Reduced States

Let's say Alice and Bob share a state $|\phi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$.

If Alice measures in base $\{|\xi_i\rangle\}$ and discard the post-measurement state, recording only the observed index i in a third register $\mathcal{H}_R$, final state is $|k\rangle_R$ w/ prob. $\sum_k \text{Tr}(\langle \xi_k | \otimes \mathbf{I}_B |\psi\rangle \langle \psi|_{AB} \langle \xi_k | \otimes \mathbf{I}_B)$, so our action corresponds to the channel

$$\mathcal{E}(\rho) = \sum_{k=0}^{\dim \mathcal{H}_A - 1} \sum_{b=0}^{\dim \mathcal{H}_B - 1} (|k\rangle_R \langle \xi_k |_A \otimes \langle b |_B) \rho (|\xi_k \rangle_A \langle k |_R \otimes |b \rangle_B).$$

We break this up into two steps: $\mathcal{E}(\rho) = \mathcal{E}_A(\text{Tr}_B(\rho))$, where

$\mathcal{E}_A(\rho_A) = \sum_{k=0}^{\dim \mathcal{H}_A - 1} |k\rangle_R \langle \xi_k |_A \rho |\xi_k \rangle_A \langle k |_R$, and we recall

$\text{Tr}_B(\rho_{AB}) = \sum_{k=0}^{\dim \mathcal{H}_B - 1} (\mathbf{I}_A \otimes \langle k |_B) \rho_{AB} (\mathbf{I}_A \otimes |k \rangle_B)$.

We conclude that in absence of any communication with Bob, Alice cannot discern whether she was holding her part of a ρ_{AB} , or she was actually holding an isolated copy of $\rho_A = \text{Tr}_B \rho_{AB}$. Reduced states play the role of marginals from the classical world.

No Communication Theorem

Interestingly, the algebra works out so that if Bob does something to his system, Alice can't find out before communicating with Bob.

Note that it is sufficient to show this for the basis elements,

$$\begin{aligned}\mathrm{Tr}_B((\mathbf{I}_A \otimes \mathcal{E}) |i_A i'_B\rangle \langle j_A j'_B|) &= \sum_k \mathrm{Tr}_B \left((\mathbf{I}_A \otimes \mathbf{K}_k) (|i_A\rangle \langle j_A| \otimes |i'_B\rangle \langle j'_B|) (\mathbf{I}_A \otimes \mathbf{K}_k^\dagger) \right) \\ &= \sum_k |i_A\rangle \langle j_A| \sum_k \mathrm{Tr} \left(\mathbf{K}_k |i'_B\rangle \langle j'_B| \mathbf{K}_k^\dagger \right) = |i_A\rangle \langle j_A| \delta_{j_B, j'_B},\end{aligned}$$

and it then follows that $\mathrm{Tr}_B((\mathbf{I}_A \otimes \mathcal{E})\rho) = \mathrm{Tr}_B \rho$ for all states ρ .

So, whatever Bob does with his subsystem – gates, measurements, entanglement with an ancilla etc – there is no measurement that Alice can perform (without assistance from Bob) that would give her any hint that Bob acted in any way. Any statistic she would compute after Bob's action would have had the same value before.

Additionally, $\mathrm{Tr}_A((\mathbf{I}_A \otimes \mathcal{E})\rho) = \mathcal{E}(\mathrm{Tr}_A \rho)$ since Bob was shown to be unable to discern whether he was holding his half of ρ or just $\mathrm{Tr}_A \rho$.

Quantum Mutual Information

Recall now that the KL divergence could be used to define classical mutual information. We thus consider the analogue definition

$$\begin{aligned} I(A : B)_\rho &= D(\rho_{AB} \| \rho_A \otimes \rho_B) \\ &= -\text{Tr}(\rho_{AB}(\log \rho_A \otimes \rho_B - \log \rho_{AB})) \\ &= -H(\rho_{AB}) - \text{Tr}(\rho_{AB}(\log \rho_A \otimes \mathbf{I}_B + \log \mathbf{I}_A \otimes \rho_B)) \\ &= -H(\rho_{AB}) - \text{Tr}(\rho_{AB} \log \rho_A \otimes \mathbf{I}_B) - \text{Tr}(\rho_{AB} \log \mathbf{I}_A \otimes \rho_B) \\ &= -H(\rho_{AB}) - \text{Tr} \text{Tr}_B(\rho_{AB} \log \rho_A \otimes \mathbf{I}_B) - \text{Tr} \text{Tr}_A(\rho_{AB} \log \mathbf{I}_A \otimes \rho_B) \\ &= -H(\rho_{AB}) - \text{Tr}(\rho_A \log \rho_A) - \text{Tr}(\rho_B \log \rho_B) \\ &= H(\rho_A) + H(\rho_B) - H(\rho_{AB}), \end{aligned}$$

as in the classical case. We used the fact that

$$\begin{aligned} &\text{Tr}_B(\rho_{AB}(\log \rho_A \otimes \mathbf{I}_B)) \\ &= \sum_{i,i',j,j'} \langle ii' | \rho | jj' \rangle \text{Tr}_B((|i\rangle \langle j|_A \otimes |i'\rangle \langle j'|_B)(\log \rho_A \otimes \mathbf{I}_B)) \\ &= \sum_{i,i',j} \langle ii' | \rho | jj' \rangle (|i\rangle \langle j| \log \rho_A) = \sum_{i,j} \langle i | \rho_A | j \rangle (|i\rangle \langle j| \log \rho_A) = \rho_A \log \rho_A. \end{aligned}$$

Example

Let's take a quantum-classical state $\rho_{AB} = \sum_i p_i |i\rangle \langle i|_A \otimes \rho_{iB}$.

To compute quantum mutual information we proceed as follows:

$$\log \rho_{AB} = \log \left(\sum_i p_i |i\rangle \langle i|_A \otimes \rho_{iB} \right) = \sum_i |i\rangle \langle i|_A \otimes \log(p_i \rho_{iB})$$

$$= \sum_i |i\rangle \langle i|_A \otimes (\log p_i \mathbf{I}_B + \log \rho_{iB})$$

$$= \sum_i \log p_i |i\rangle \langle i|_A \otimes \mathbf{I}_B + \sum_i |i\rangle \langle i|_A \otimes \log \rho_{iB},$$

$$\rho_{AB} \log \rho_{AB} = \sum_i p_i \log p_i |i\rangle \langle i|_A \otimes \rho_{iB} + \sum_i p_i |i\rangle \langle i|_A \otimes \rho_{iB} \log \rho_{iB}$$

$$H(\rho_{AB}) = -\text{Tr} \rho_{AB} \log \rho_{AB} = -\sum_i p_i \log p_i - \sum_i p_i \text{Tr} \rho_i \log \rho_i$$

$$I(A : B)_\rho = H(\rho_B) + H(\rho_A) - H(\rho_{AB})$$

$$= H\left(\sum_i p_i \rho_i\right) + \underbrace{H(\mathbf{p}) - H(\mathbf{p})}_0 - \sum_i p_i H(\rho_i).$$

Holevo Theorem

Now note that if we take part of a bipartite state ρ_{AB} through a channel $\mathcal{E}$ we had $((\mathbf{I}_A \otimes \mathcal{E})\rho)_A = \rho_A$ and $((\mathbf{I}_A \otimes \mathcal{E})\rho)_B = \mathcal{E}(\rho_B)$ so

$$\begin{aligned} I(A : \mathcal{E}(B))_\rho &= I(A : B)_{\mathbf{I} \otimes \mathcal{E}(\rho)} = D(\mathbf{I} \otimes \mathcal{E}(\rho) \| \rho_A \otimes \mathcal{E}(\rho_B)) \\ &\leq D(\rho \| \rho_A \otimes \rho_B) = I(A : B)_\rho, \end{aligned}$$

known as the monotonicity of quantum mutual information.

In particular, if Alice prepares a quantum-classical state and sends the quantum part to Bob, any measurement he could make, including POVMs, ultimately takes the form of a such a channel,

$$\rho = \sum_i p_i |i\rangle \langle i|_A \otimes \rho_{iB} \mapsto \sum_{i,j} p_i |i\rangle \langle i|_A \otimes d_{j,i} |j\rangle \langle j|_B = \mathbf{I}_A \otimes \mathcal{E}(\rho),$$

so the best approximation that Bob can make of Alice's initial distribution has its quality bounded by

$$I(A : \tilde{A}) = I(A : \mathcal{E}(B))_\rho \leq I(A : B)_\rho = H\left(\sum_i p_i \rho_i\right) - \sum_i p_i H(\rho_i).$$

Loopholes

Let's say Alice wants to send to Bob n classical dits using 1 qudit. Without entanglement, best she can do is send Bob a mixture of d^n mixed states and hope he has a clever measurement at hand. Notice this is exactly the scenario in Holevo's theorem. And thus, in this scenario, Bob can only decode 1 dit.

The first loophole is that the theorem doesn't say which bit. More specifically, it doesn't say he can't choose the bit he gets. This is the loophole powering Quantum Random Access Codes.

The next loophole is what happens if we had entanglement. This is the loophole powering Superdense Coding. If instead of preparing ρ_i Alice applies $\mathcal{F}_i$ to her half of a bipartite state $\sigma_{A'B}$ then it can be shown that the limit increases to

$$I(A : \tilde{A}) \leq \log d - H(A' | B)_\sigma,$$

where the conditional entropy $H(A' | B)_\sigma = H(\sigma_{A'B}) - H(\sigma_B)$ can be as negative as $H(A' | B)_{|\beta_{00}\rangle\langle\beta_{00}|} = -\log d$.

Conditional von Neumann Entropy

The conditional von Neumann entropy can also be computed as

$$\begin{aligned} H(A | B)_\rho &= H(\rho_{AB}) - H(\rho_B) \\ &= -\text{Tr } \rho_{AB} \log \rho_{AB} + \text{Tr } \rho_B \log \rho_B \\ &= -\text{Tr } \rho_{AB} \log \rho_{AB} + \text{Tr} \sum_k (\langle k| \otimes \mathbf{I}) \rho_{AB} (|k\rangle \otimes \mathbf{I}) \log \rho_B \\ &= -\text{Tr } \rho_{AB} \log \rho_{AB} + \text{Tr } \rho_{AB} \sum_k (|k\rangle \otimes \mathbf{I}) \log \rho_B (\langle k| \otimes \mathbf{I}) \\ &= -\text{Tr } \rho_{AB} \log \rho_{AB} + \text{Tr } \rho_{AB} \log \mathbf{I}_A \otimes \rho_B \\ &= -D(\rho_{AB} \| \mathbf{I}_A \otimes \rho_B) = \log d_A - D(\rho_{AB} \| \frac{\mathbf{I}_A}{d_A} \otimes \rho_B), \end{aligned}$$

so by the Data Processing Inequality and Joint Convexity,

1. $H(A | \mathcal{E}(B))_\rho \geq H(A | B)_\rho$ for any channel $\mathcal{E}$;
2. $H(\mathcal{E}(A) | B)_\rho \geq H(A | B)_\rho$ for any bistochastic channel $\mathcal{E}$;
3. For separable ρ i.e. $\rho = \sum_i p_i |\phi_i\rangle \langle \phi_i| \otimes |\psi_i\rangle \langle \psi_i|$ we have

$$\begin{aligned} H(A | B)_\rho &= \log d_A - D\left(\sum_i p_i |\phi_i\rangle \langle \phi_i| \otimes |\psi_i\rangle \langle \psi_i| \parallel \sum_i p_i \frac{\mathbf{I}_A}{d_A} \otimes |\psi_i\rangle \langle \psi_i|\right) \\ &\geq \log d_A - \sum_i p_i D(|\phi_i\rangle \langle \phi_i| \otimes |\psi_i\rangle \langle \psi_i| \parallel \frac{\mathbf{I}_A}{d_A} \otimes |\psi_i\rangle \langle \psi_i|) = 0. \end{aligned}$$

Schmidt Decompositions & Entropy of Entanglement

Let's look at what happens for pure states $\rho_{AB} = |\psi\rangle\langle\psi|$. Then, $H(\rho_{AB}) = 0$ so $H(A | B)_\rho = -H(\rho_B)$ and $H(B | A)_\rho = -H(\rho_A)$, so both are zero iff $|\psi\rangle$ is separable and negative otherwise.

However, for any $|\psi\rangle \in \mathbb{C}^d \otimes \mathbb{C}^{d'}$, say $|\psi\rangle = \sum_{i,j} \alpha_{i,j} |i\rangle \otimes |j\rangle$, we have the singular value decomposition of the amplitude matrix

$$\begin{bmatrix} \alpha_{0,0} & \dots & \alpha_{0,d'-1} \\ \vdots & \ddots & \vdots \\ \alpha_{d,0} & \dots & \alpha_{d,d'-1} \end{bmatrix} = \mathbf{M} = \mathbf{U} \mathbf{S} \mathbf{V}^\dagger = \sum_{k=0}^{\text{rank } \mathbf{M}-1} |\phi_k\rangle s_k \langle \xi_k|,$$

where $\{|\phi_i\rangle\}$ form a basis of $\mathbb{C}^d$, $\{|\xi_j\rangle\}$ form a basis of $\mathbb{C}^{d'}$ and $\alpha_{i,j} = \langle i | \mathbf{M} | j \rangle = \sum_k \langle i | \phi_k \rangle s_k \langle \xi_k | j \rangle$ so $|\psi\rangle = \sum_k s_k |\phi_k\rangle \otimes |\overline{\xi_k}\rangle$, thus the reduced states are $\sum_k s_k |\phi_k\rangle \langle \phi_k|$ and $\sum_k s_k |\overline{\xi_k}\rangle \langle \overline{\xi_k}|$, so they have the same entropy, **the entanglement entropy of $|\psi\rangle$** .

The trick above is called **Schmidt decomposition** of $|\psi\rangle$. The s_k are called Schmidt coefficients and $\text{rank } \mathbf{M}$ is $|\psi\rangle$'s **Schmidt rank**.

Uncertainty in the presence of Quantum Memory

Since $H(\mathcal{E}(A) | B)_\rho \geq H(A | B)_\rho$ for any bistochastic channel $\mathcal{E}$, the conditional von Neumann entropy acts as a sort of uncertainty measure about Alice's system, given the state of Bob's system.

In particular, if Alice measures observable $\mathbf{O} = \sum_k \lambda_k |\phi_k\rangle \langle \phi_k|$, the conditional uncertainty about Alice's measurement given the state of Bob's system would be

$$H(\mathbf{O} | B)_\rho = H(\mathcal{E}(A) | B)_\rho = H(A | B)_{(\mathcal{E} \otimes \mathbf{I}_B)(\rho)}.$$

This concept allowed Berta et al. to improve the Maassen-Uffink uncertainty relation to account for quantum memory,

$$H(\mathbf{O} | B)_\rho + H(\mathbf{O}' | B)_\rho \geq -\log \max_{k,j} \underbrace{|\langle \phi_k | \xi_j \rangle|^2}_{\text{overlap}} + H(A | B)_\rho,$$

and if there is no quantum memory the left term becomes classical,

$$H(\mathbf{O}) + H(\mathbf{O}') \geq -\log \max_{k,j} |\langle \phi_k | \xi_j \rangle|^2 + \underbrace{H(\rho)}_{\geq 0}.$$

Tripartite Uncertainty Relations

However, we know that for any state $\rho_{AB} = \sum_i p_i |\psi_i\rangle \langle \psi_i|_{AB}$ there is $|\phi\rangle_{ABC} = \sum_i \sqrt{p_i} |\psi_i\rangle_{AB} \otimes |\psi_i\rangle_C$ s.t. $\rho_{AB} = \text{Tr}_C |\phi\rangle \langle \phi|_{ABC}$ so

$$\begin{aligned} H(A | B)_{\rho_{AB}} &= H(\rho_{AB}) - H(\rho_B) = H(\rho_C) - H(\rho_B) \\ H(O' | B)_{\rho_{AB}} &= H(A | B)_{(\mathcal{E} \otimes \mathbf{I}_B)\rho_{AB}} \\ &= H((\mathcal{E} \otimes \mathbf{I}_B)\rho_{AB}) - H(\rho_B) \\ &= H((\mathcal{E} \otimes \mathbf{I}_B)\rho_{AB}) - H(\rho_C) + H(A | B) \\ &\leq H((\mathcal{E} \otimes \mathbf{I}_C)\rho_{AC}) + H(A | B), \end{aligned}$$

intuitively because if we measure C we get i and so if we also get Alice's result then we will know Bob's post-measurement state.

We conclude that for pure states we have

$$H(O | B)_\rho + H(O' | C)_\rho \geq -\log \max_{k,j} |\langle \phi_k | \xi_j \rangle|^2,$$

and so if we average the left hand side we will find this inequality also holds for mixed states.

Quantum Key Distribution: BB84 Protocol

Let's say Alice sends Bob a qubit in random state $|\psi\rangle = \mathbf{H}^i \mathbf{X}^j |0\rangle$.

If Bob guesses i he can measure observable $\mathbf{H}^i \mathbf{Z} \mathbf{H}^i$ to get j .

If he doesn't, the round is discarded and the protocol restarted.

To attack this protocol, Eve needs to intercept $|\psi\rangle$ in flight.

We can model this by applying a channel $\mathcal{E}$ to the in-flight $|\psi\rangle$ and another system on the side, whether she entangles them or not.

However, notice that $|\psi\rangle$ could have been obtained for example by measuring either half of a Bell state. In particular, if it was Alice's state that was measured, this measurement would commute with the application of $\mathcal{E}$. We find the equivalent protocol:

1. Alice prepares state $|\beta_{00}\rangle_{AB}$;
2. Alice sends half of this state to Eve;
3. Eve applies $\mathcal{E}$ to the in-flight system B and her system E ;
4. Eve sends the in-flight system B to Bob;
5. Alice and Bob measure in random bases;
6. If they measured the same thing they keep the result.

Quantum Key Distribution: BB84 Protocol

After Eve's intervention Bob gets part of the state ρ_{ABE} so

$$H(\mathbf{X} \mid \mathbf{B})_{\rho} + H(\mathbf{Z} \mid \mathbf{E})_{\rho} \geq 1, \quad H(\mathbf{Z} \mid \mathbf{B})_{\rho} + H(\mathbf{X} \mid \mathbf{E})_{\rho} \geq 1,$$

thus if Bob's error rate measuring $\mathbf{X}$, δ_{bit} , is low this proves Eve's uncertainty about $\mathbf{Z}$ is high, $1 - h(\delta_{bit})$. Similarly, if Bob's error rate measuring $\mathbf{Z}$, δ_{phase} , is low this proves Eve's uncertainty about $\mathbf{X}$ is high, $1 - h(\delta_{phase})$. On average, if the probability that Eve can guess j is p then $h(p) \geq 1 - \frac{1}{2}h(\delta_{bit}) - \frac{1}{2}h(\delta_{phase})$ and so

$$p \leq \frac{1}{2}(1 + \sqrt{(2 \ln 2)(1 - h(p))}) \leq \frac{1}{2}(1 + \sqrt{\ln 2(h(\delta_{bit}) + h(\delta_{phase}))}),$$

e.g. if the error rate is 1% then Eve can guess less than 66.6% bits.

If we use QKD to share 256-bit keys, the 1% mismatch rate means there is a 10^{-25} chance that there are > 32 errors. But, if we send 32 parity sums Eve gets 0.125 bits per round so $h(p) \geq 0.7942$, $p \leq 76.06\%$ and so she has probability of 10^{-23} to share their key.

Follow-up Reading

- ▶ J.Watrous – The theory of quantum information
Cambridge university press (2018)
5.2.3: Proof of Joint-Convexity of Umegaki Divergence
- ▶ P.Coles et al – Entropic uncertainty relations and their applications
Reviews of Modern Physics 89.1 (2017): 015002
Appendix C3: Proof of the Berta et al. tripartite EUR